

Aircraft Fault Analysis Report

Report generated on 2026-04-25 23:48. The workflow rebuilds the aircraft dataset from raw Excel files, applies healthy-reference anomaly detection, infers provisional fault onsets, trains grouped fault-prediction models, and reports both row-level and file-level reliability.

The analysis avoids row-level leakage by grouping validation by file. Direct anomaly-score columns are excluded from the supervised prediction features so that the downstream model does not simply relearn the anomaly detector.

The current fault-onset labels remain provisional. They are inferred from sustained anomaly behavior and should be reviewed against engineering logs or manually annotated onset points before strong scientific claims are made.

Fault prediction metrics

	metric	value
	model	hist_gradient_boosting
	rows	430581
	files	36
	roc_auc	0.9999999693168723
average_precision		0.9999973274119679
precision		0.9979633401221996
recall		1.0
f1		0.9989806320081549
balanced_accuracy		0.9999882541151708
selected_threshold		0.6
healthy_file_false_alarm_rate		0.0
defective_file_detection_rate		1.0

Anomaly quality metrics

	metric	value
	selected_threshold	1.911752700805664
	file_rate_threshold	0.03
healthy_file_false_alarm_rate		0.0909090909090909
defective_file_detection_rate		1.0
healthy_row_false_alarm_rate		0.015003996734823
defective_row_detection_rate		0.0636411733584151
	row_roc_auc	0.6633997987568214
	objective	0.7087589627096393
	w_pca	0.0
	w_iso	1.0
	w_maha	0.0

Methodology Summary

The anomaly stage uses robust scaling fitted only on healthy-reference data, PCA reconstruction error, Isolation Forest scoring, and a Mahalanobis-style score in PCA space. The final anomaly score is selected from a grid of ensemble weights and thresholds using file-level detection and false-alarm penalties.

The prediction stage creates causal temporal features only from current and previous rows within each file. Features include current values, first differences, rolling means, rolling standard deviations, rolling slopes, exponentially weighted means, and missingness indicators.

Candidate models are compared using grouped cross-validation. The selected model is then refit and evaluated with precision-recall, ROC, calibration, confusion matrix, file-level prediction scores, feature importance, permutation importance, and timeline plots.

Model comparison

	model	valid_rows	roc_auc	average_precision	precision	recall	f1	balanced_accuracy
hist_gradient_boosting	random_forest	82342	0.9999997549186251	0.9999961972308548	0.9997959183673469	0.9997959183673469	0.9997959183673469	0.9998915027388502
	extra_trees	82342	0.9997455132979046	0.996408788987792	0.9837330552659019	0.9626530612244898	0.9730789066529139	0.9808229279160335
	logistic_robust	82342	0.999499607079215	0.9924171950767804	0.9458762886597938	0.9736734693877551	0.9595736122284795	0.9850741252571378
		82342	0.9875946844221534	0.7636280904254048	0.6581327396688466	0.9653061224489796	0.7826590551832547	0.9667895762938321

Fault onset reference

onset_position	onset_sequence_index	method	confidence	pre_onset_median	post_onset_median	file_name	group
1939	1939	sustained_threshold_crossing	low	1.501918911933899	0.7197118997573853	Defect15(01).xlsx	defective
1698	1698	sustained_threshold_crossing	low	1.6252636909484863	1.3488874435424805	Defect15(02).xlsx	defective
15639	15639	sustained_threshold_crossing	medium	0.080622524023056	2.001061201095581	Defect15(03).xlsx	defective
1312	1312	sustained_threshold_crossing	low	1.906025528907776	0.2988683581352234	Defect15(04).xlsx	defective
1435	1435	sustained_threshold_crossing	low	1.603426814079285	0.7091450095176697	Defect16(01).xlsx	defective
1587	1587	sustained_threshold_crossing	low	1.177343249320984	0.2322175353765487	Defect16(02).xlsx	defective
14661	14661	sustained_threshold_crossing	medium	0.0110667096450924	1.9446580410003664	Defect16(03).xlsx	defective
1939	1939	sustained_threshold_crossing	low	1.501918911933899	0.7197118997573853	Defected 1.xlsx	defective
1698	1698	sustained_threshold_crossing	low	1.6252636909484863	1.3488874435424805	Defected 2.xlsx	defective
15639	15639	sustained_threshold_crossing	medium	0.080622524023056	2.001061201095581	Defected 3.xlsx	defective
1312	1312	sustained_threshold_crossing	low	1.906025528907776	0.2988683581352234	Defected 4.xlsx	defective
1435	1435	sustained_threshold_crossing	low	1.603426814079285	0.7091450095176697	Defected 5.xlsx	defective

Reliability And Audit Notes

The main reliability risk is still the quality of the inferred pre-fault target. External onset labels would make the results more scientifically defensible.

The file-level false alarm rate should be treated as a deployment gate. A model with high row-level ROC-AUC can still be unsuitable if healthy files repeatedly cross the alert threshold.

The leakage audit records whether any engineered feature name contains blocked target, anomaly, fault, or label terms. The model feature construction intentionally uses raw sensor-derived columns rather than direct anomaly scores.

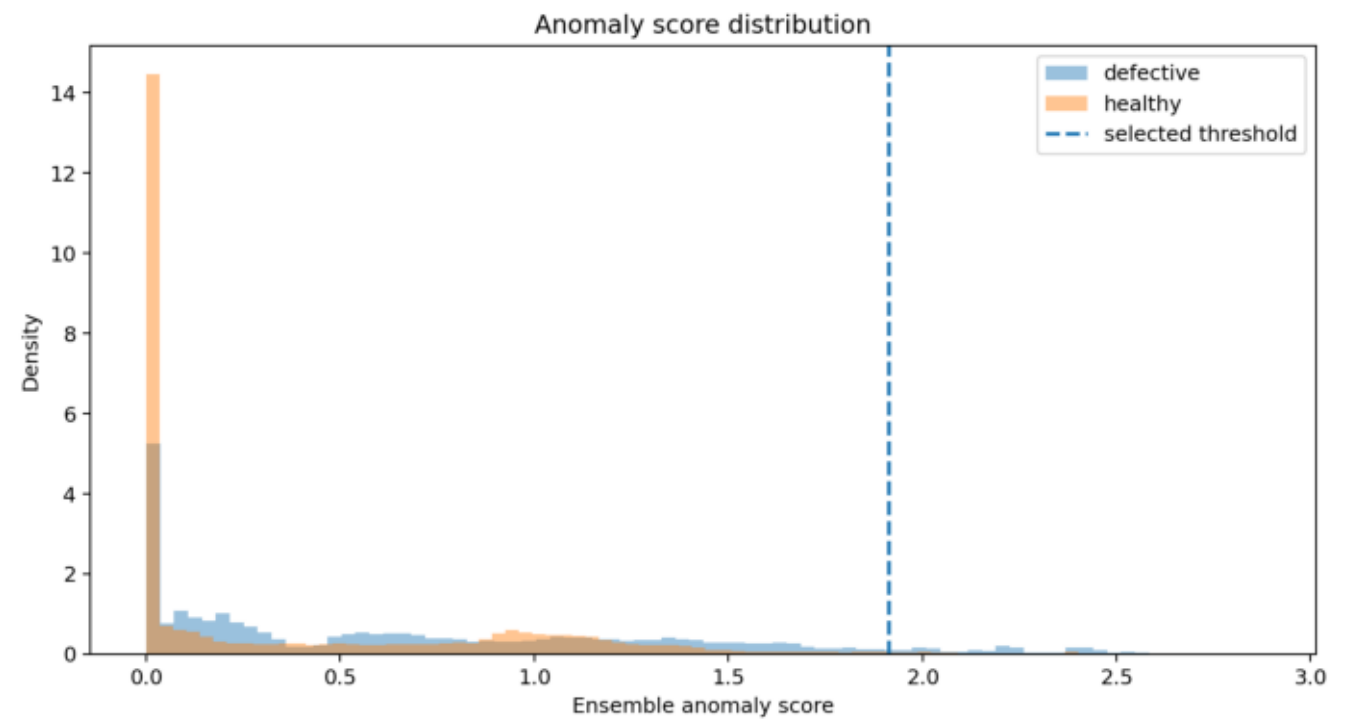
Reliability flags

provisional_onset_labels	flag severity		detail
	medium	Onsets are inferred from anomaly behavior and should be reviewed	manua

File prediction scores

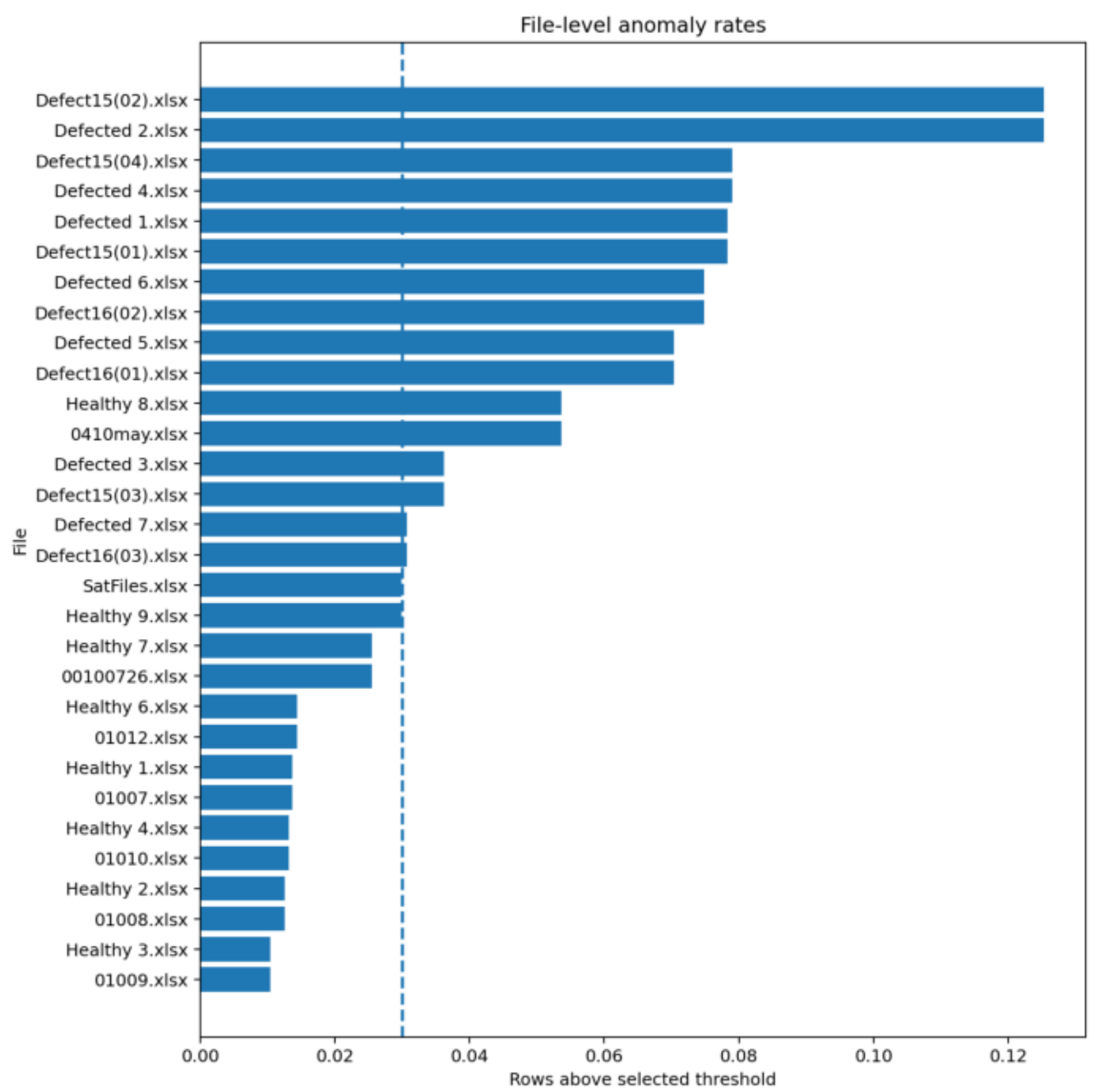
file_name	group	rows	max_prediction_score	mean_prediction_score	predicted_rows	actual_positive_rows	predicted_positive_rate
Defected 5.xlsx	defective	1435	0.99967676	0.24381311	350	350	0.24390243902439024
Defect16(01).xlsx	defective	1435	0.99967676	0.24381311	350	350	0.24390243902439024
Defect15(03).xlsx	defective	15639	0.9996762	0.022398818	350	350	0.022379947566979985
Defect16(03).xlsx	defective	14661	0.9996762	0.024226224	355	350	0.024213900825318872
Defected 3.xlsx	defective	15639	0.9996762	0.022398818	350	350	0.022379947566979985
Defected 7.xlsx	defective	14661	0.9996762	0.024226224	355	350	0.024213900825318872
Defected 4.xlsx	defective	1312	0.9996727	0.26670593	350	350	0.26676829268292684
Defect15(04).xlsx	defective	1312	0.9996727	0.26670593	350	350	0.26676829268292684
Defect15(02).xlsx	defective	1698	0.9996624	0.20606978	350	350	0.2061248527679623
Defected 2.xlsx	defective	1698	0.9996624	0.20606978	350	350	0.2061248527679623
Defected 1.xlsx	defective	1939	0.99963784	0.18045755	350	350	0.18050541516245489
Defect15(01).xlsx	defective	1939	0.99963784	0.18045755	350	350	0.18050541516245489

Anomaly score distribution



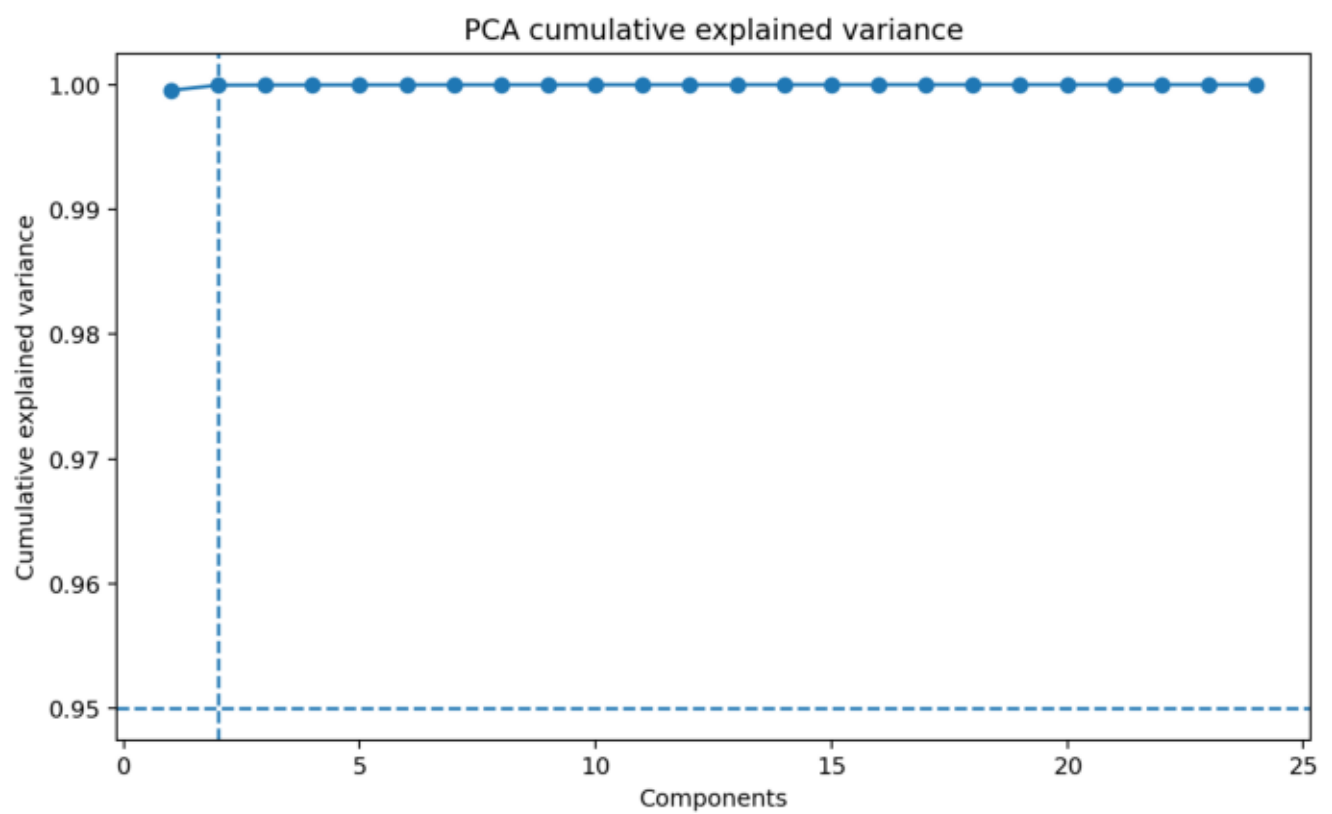
This plot compares anomaly-score density for healthy and defective files and shows the selected alert threshold.

File-level anomaly rates



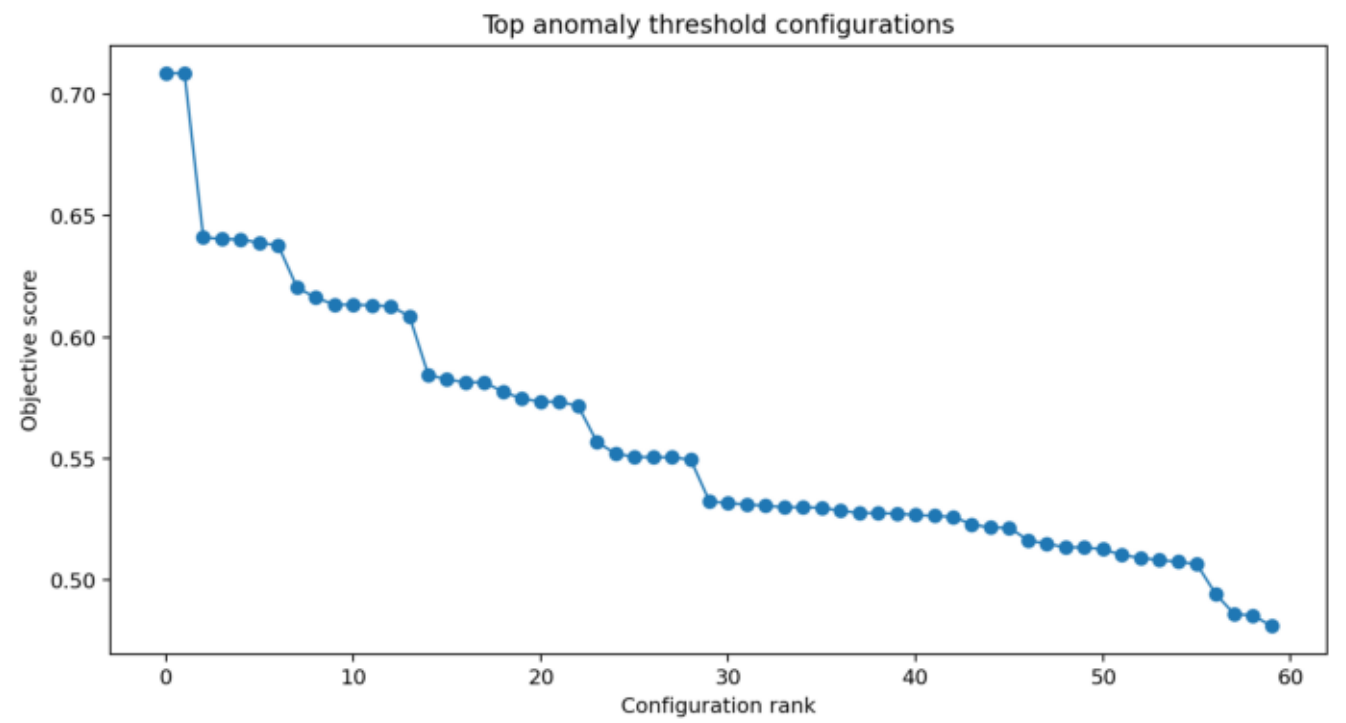
This chart shows how frequently each file crosses the selected anomaly threshold.

PCA explained variance



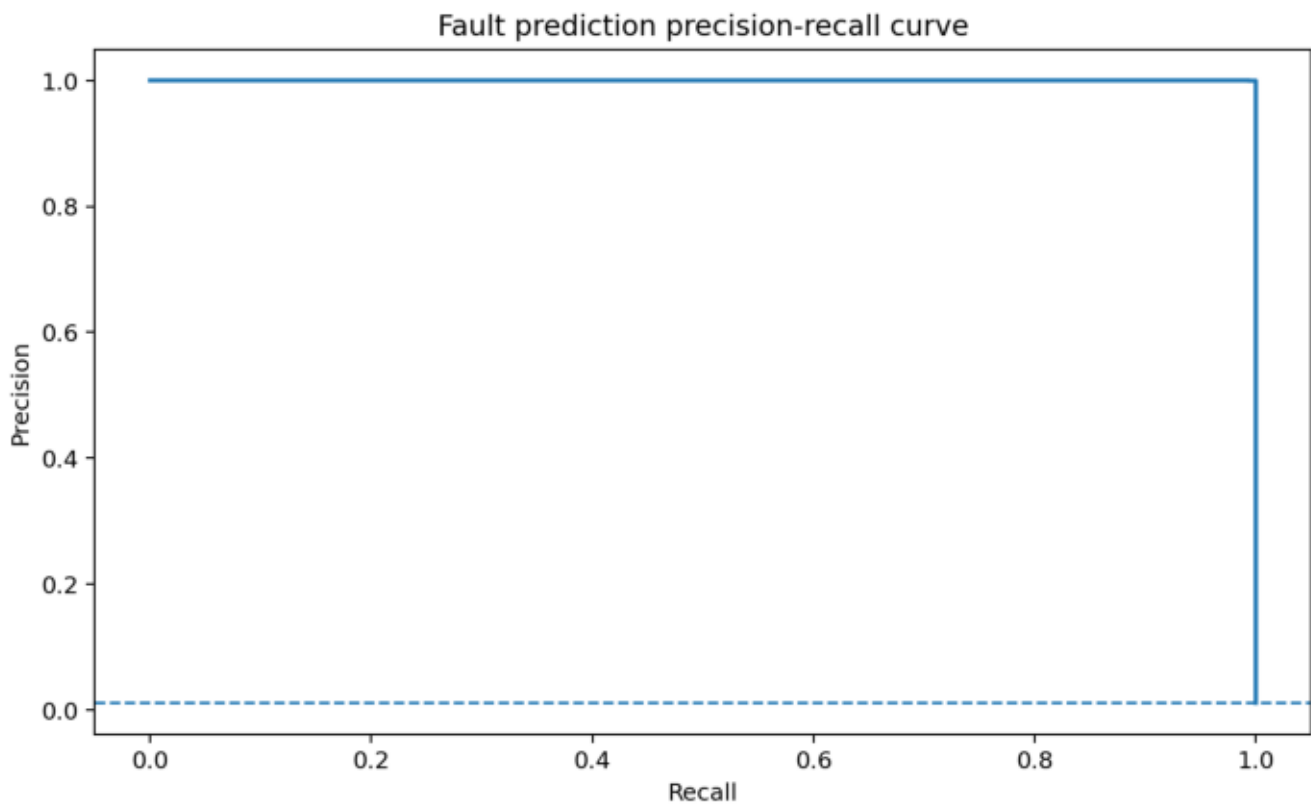
This figure shows the cumulative PCA variance used in the reconstruction-error model.

Anomaly threshold search



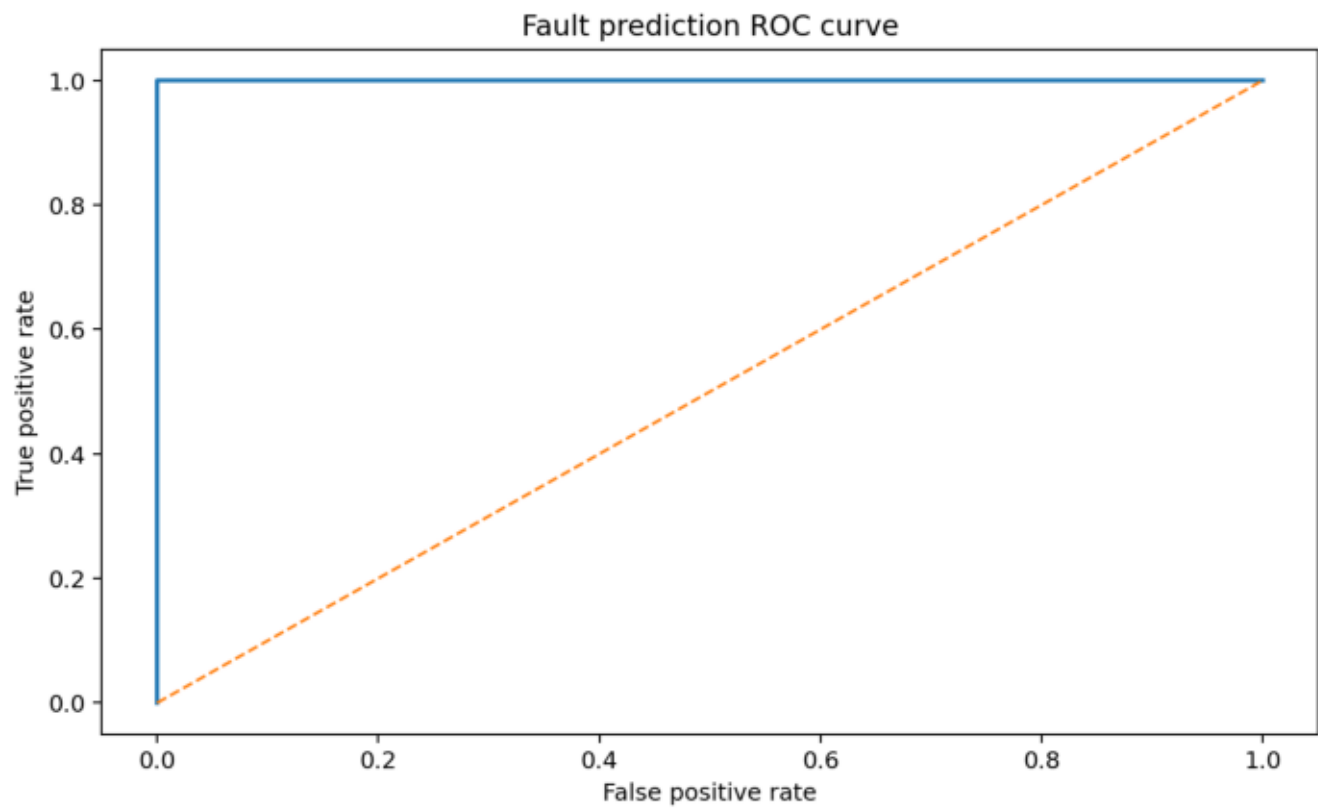
This plot ranks threshold configurations by the validation objective.

Fault prediction precision-recall curve



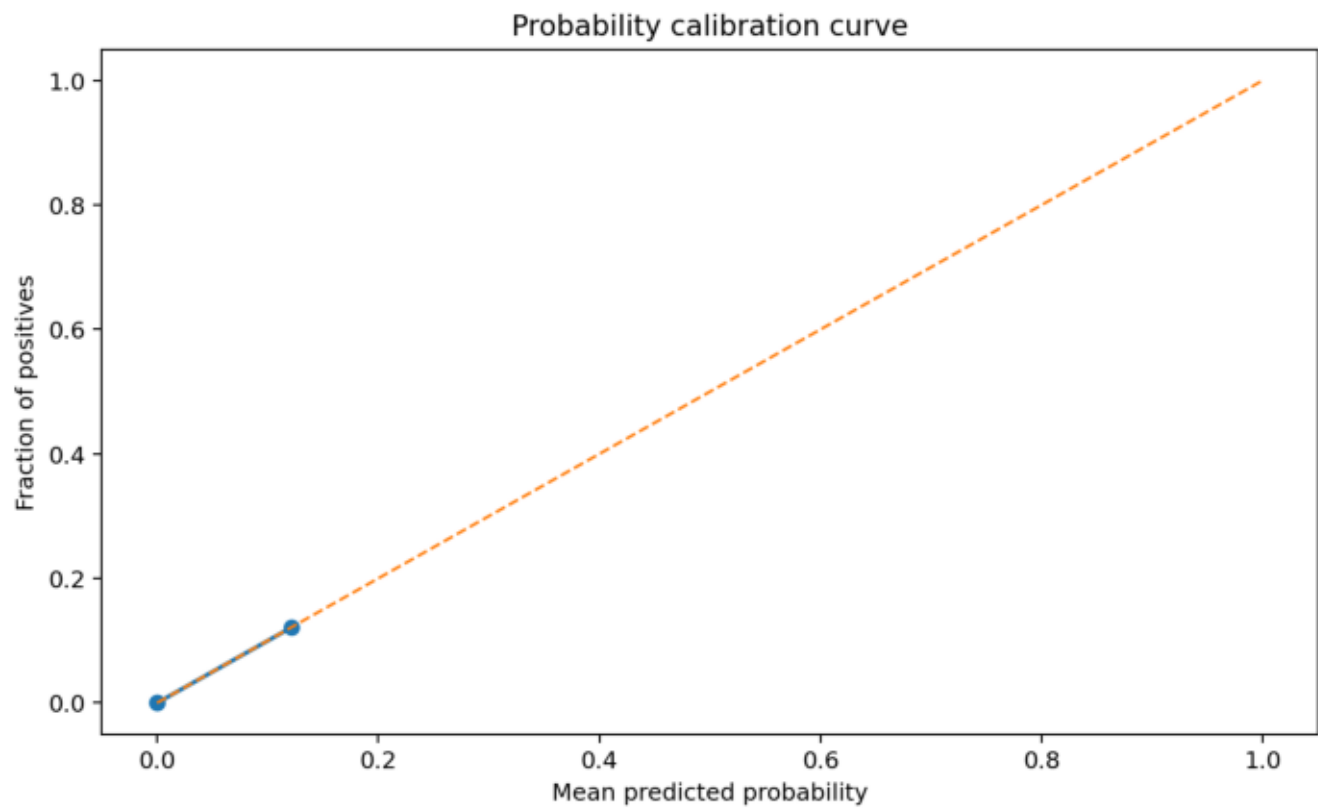
This curve is more informative than accuracy for rare pre-fault labels.

Fault prediction ROC curve



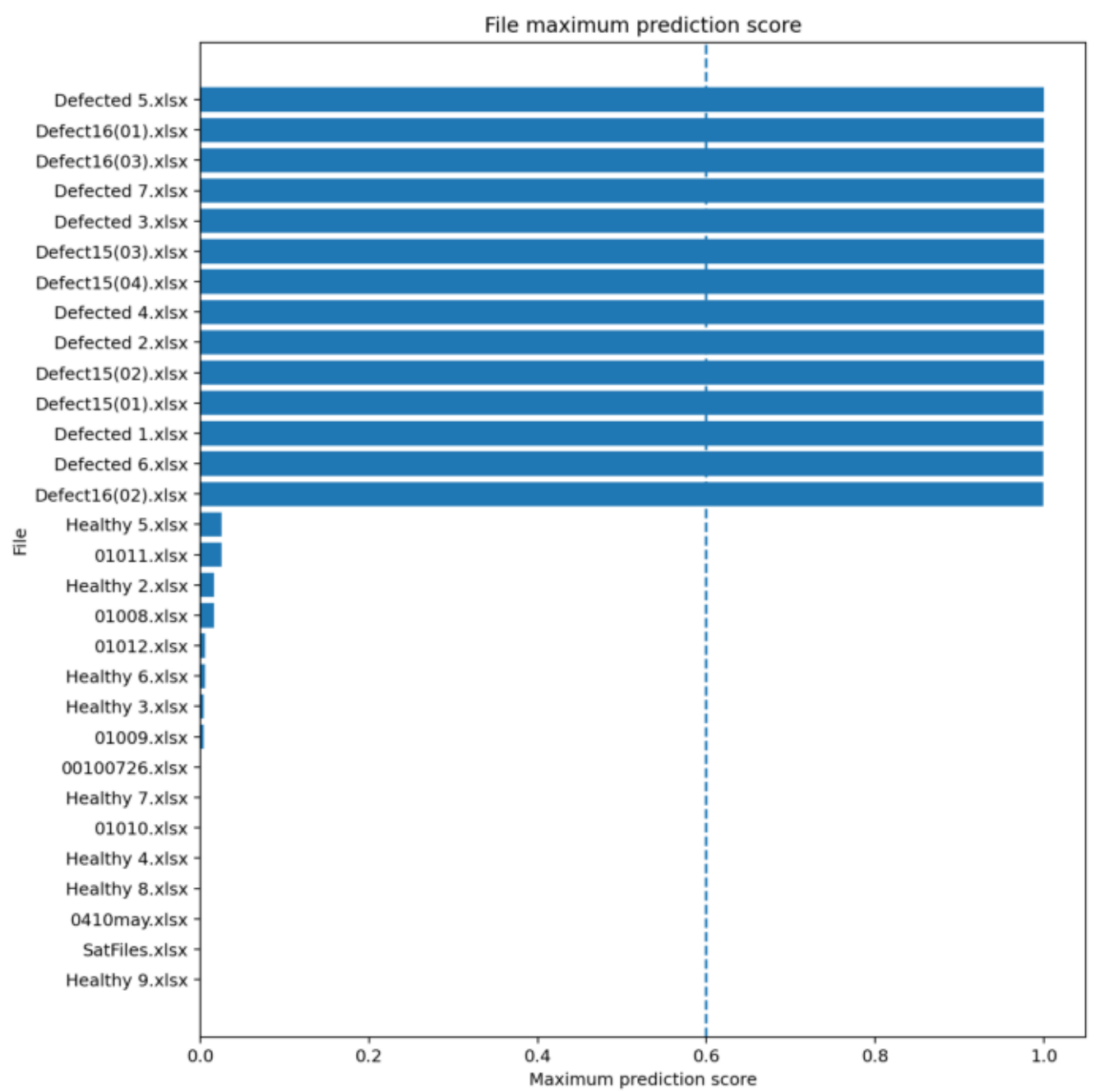
This curve summarizes separability between normal and provisional pre-fault rows.

Probability calibration



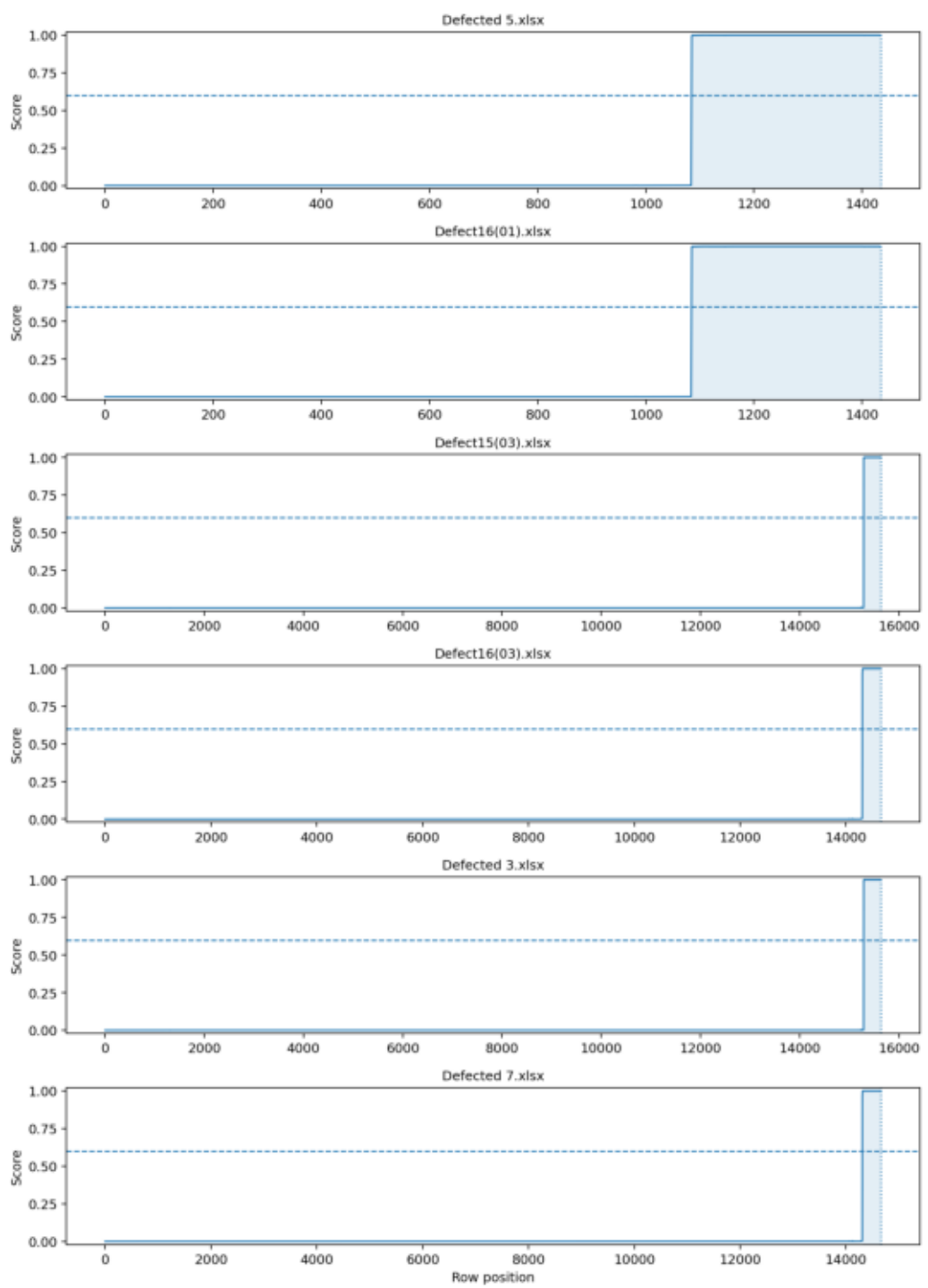
This plot checks whether predicted probabilities are numerically reliable.

File-level prediction scores



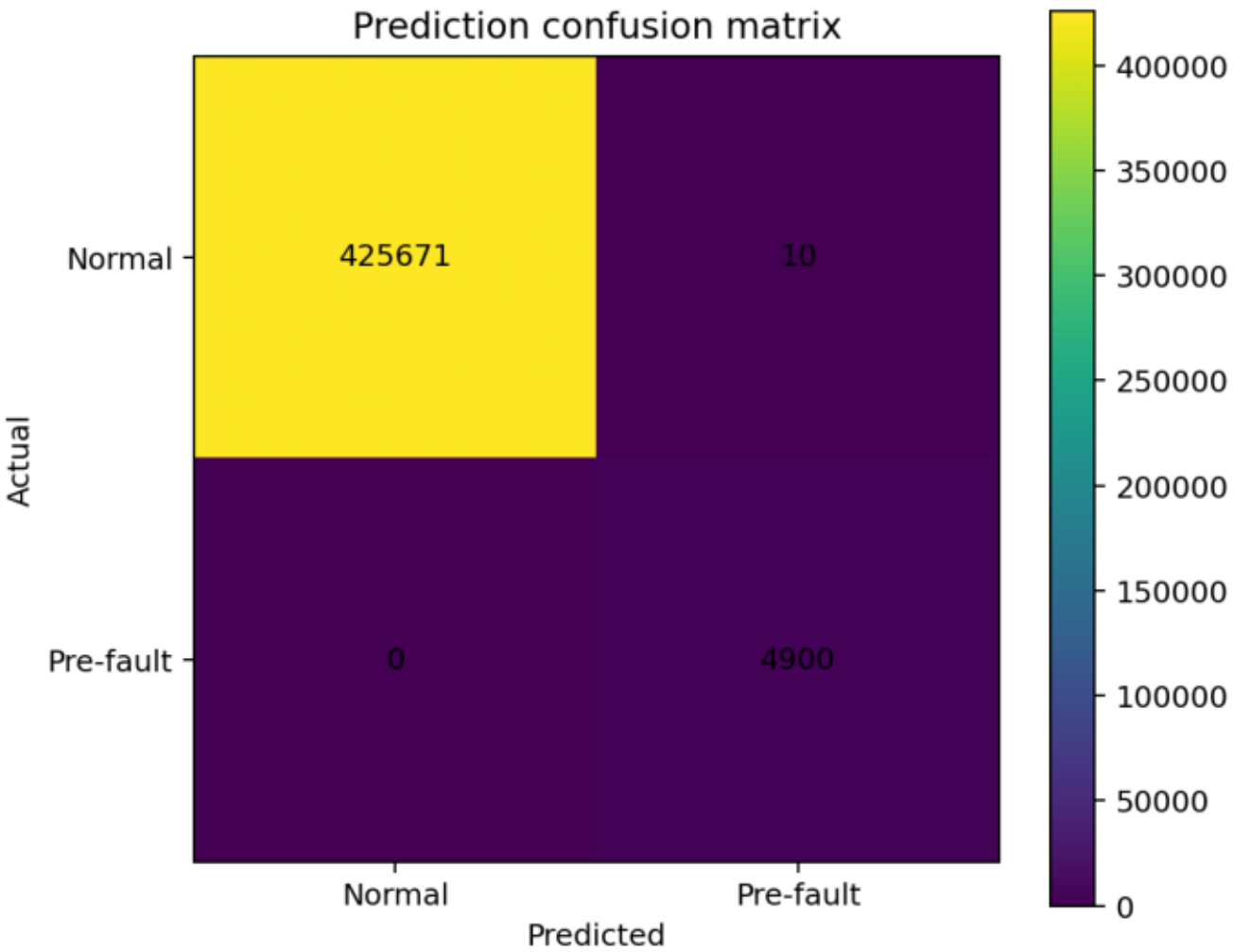
This chart supports deployment review by showing which files cross the selected prediction threshold.

Prediction timelines



These timelines show how prediction scores evolve within the highest-risk files.

Confusion matrix



The matrix summarizes row-level decisions at the selected threshold.